

Newton

+91 9930742189

newton4050@gmail.com

www.linkedin.com/in/newton-computer-architect

Current Position

2026–Present **Assistant Professor, BITS Pilani, K K Birla Goa Campus, Goa, India.**

Professional Skills

Performance Workload Characterization, Performance modeling of a hardware architecture, Re-
Modeling of search and Development of memory subsystem of processor with a focus on the
Processors cache hierarchy

Research Interest

Computer Design and evaluation of secure, energy-efficient, and high performance architecture
Architecture of future computers. Modern and emerging memories (3-D stacked memories, Nov-volatile memories). Management of on-chip caches, and preventing them from side-channel attacks. Hardware accelerators for domain-specific applications like graph analytics, deep neural networks

Education

2015–2022 **Ph.D. in Computer Architecture, Indian Institute of Technology (IIT) Bombay & National University of Singapore (NUS), CGPA 9.14/10.**

2011–2014 **M.Tech. in Electrical Engineering, IIT Bombay, CGPA 9.3/10.**

2004–2008 **B.Tech. in Electronics and Communication Engineering, National Institute of Technology (NIT) Kurukshetra, India, CGPA 8.2/10.**

Professional Experience

Jun 2022–Jun **Silicon SoC Architect, Intel Technology India Pvt. Ltd., India.**

2026 Research and development of microarchitectural features in the cache hierarchy of latest intel processors

Research Experience

Oct 2021–Jun **Graduate Intern, Intel Technology India Pvt. Ltd..**

2022 Exploring and modeling new architectural techniques for memory compression predictors

2015–2023 **Graduate Research Assistant, IIT Bombay and NUS.**

Advisors: Prof. Virendra Singh, Prof. Trevor E. Carlson

Thesis Title: *Improving Performance of Modern Memory Subsystem*

- Designed energy-efficient hardware accelerators for graph analytics using the **processing-in-memory (PIM) technique and 3-D stacked memories**.
- Modeled 3-D stacked memories like Micron's Hybrid Memory Cube (HMC) or High Bandwidth Memory (HBM) to study their efficacy in enhancing performance and energy-efficiency of graph analytics.
- Exploited **Hybrid caches comprising of non-volatile STTRAM- and SRAM-based cache lines** to enhance the performance and energy-efficiency of SPEC workloads.
- Designed novel **cache replacement policies** to efficiently manage last-level caches.
- Used Sniper architectural simulator and booksim network simulator to explore and evaluate novel hardware architectures.

- 2013–2014 **Master's Thesis**, *Department of Electrical Engineering (EE), IIT Bombay.*
Advisor: Prof. Virendra Singh
Title: *Scheduling Heterogeneous Multi-Cores: Using Performance Variation Across Phases*
 Categorized applications' execution time into memory-intensive and compute-intensive phases. Proposed a dynamic scheduler to use the above categorization for executing a given application's phase on a suitable core in the heterogeneous multi-core system.
- 2015–2022 **Teaching Assistant**, *Department of Electrical Engineering, IIT Bombay.*
 Assisting in designing and evaluating course assignments and projects in the area of computer architecture, digital circuits, analog devices, and microprocessors.
- 2011–2014 **Research Assistant**, *Wadhwani Electronics Lab, EE Department, IIT Bombay.*
 Designed course experiments and assisted students in performing them. Evaluated course projects in the area of digital circuits, analog devices, and microprocessors.

Peer-Reviewed Publications

- ISVLSI'26 **"RAGE: A Data-Aware Aggressive Prefetcher for High-Performance Graph Analytics"**, Varun Venkitaraman, Mitul Tandon, Tejeshwar Bhagatsing Thorawade, Nayan Barhate, Keerthi Sagar Kokkiligadda, Newton, Raj Kumar Choudhary, Virendra Singh, IEEE Computer Society Annual Symposium on VLSI ISVLSI 2026, Kolkata, India, 2026.
- VDAT'25 **"Spin-TLB: STT-RAM-based Translation Hierarchy for Server Applications in CPUs"**, T. B. Thorawade, K. Kokkiligadda, V. Venkitaraman, Ankith, Newton, and V. Singh, The 29th International Symposium on VLSI Design and Test, Chandigarh, India, 2025.
- DATE'22 **"Data-Aware Cache Management for Graph Analytics"**, Neelam Sharma*, Varun Venkitaraman*, Newton*, Vikash Kumar, Chandan Jha, Design Automation and Test in Europe Conference, 2022, *Equal Contribution, Core-Ranking 'B'.
- CAL'21 **"Effective Migration Techniques for STT-RAM-based Hybrid Caches"**, Arindam Sarkar*, Newton*, Varun Venkitaraman, and Virendra Singh, IEEE Computer Architecture Letters, *Equal Contribution, Impact Factor: 1.4.
- CAL'20 **"PIM-GraphSCC: PIM-based Graph Processing using Graph's Community Structure"**, Newton, Virendra Singh, and Trevor E. Carlson, IEEE Computer Architecture Letters 2020, Impact Factor: 1.4.
- ICCD'19 **"Freeflow Core: Enhancing Performance of In-Order Cores with Energy Efficiency"**, Raj Kumar Choudhary, Newton, Harideep Nair, Rishabh Rawat, and Virendra Singh, The 37th IEEE International Conference on Computer Design, Abu Dhabi, United Arab Emirates, November 2019, Core-Ranking 'B'.
- GLSVLSI'18 **"AB-Aware: Application Behavior Aware Management of Shared Last Level Caches"**, Suhit Pai, Newton, and Virendra Singh, ACM Great Lakes Symposium on VLSI (GLSVLSI) 2018, Chicago, Illinois, USA, May 2018.
- ICCD'17 **"DAAIP: Deadblock Aware Adaptive Insertion Policy for High Performance Caching"**, Newton, Sujit Kumar Mahto, Suhit Pai, and Virendra Singh, The 35th IEEE International Conference on Computer Design, Boston, USA, November 2017, Core-Ranking 'B'.

VDAT'17 **"ACAM: Application Aware Adaptive Cache Management for Shared LLC"**, *Sujit Kumar Mahto and Newton*, The 21st International Symposium on VLSI Design and Test, Roorkee, India, July 2017.

GLSVLSI'17 **"EEAL: Processors' Performance Enhancement Through Early Execution of Aliased Loads"**, *Abhishek Rajgadia, Newton and Virendra Singh*, ACM Great Lakes Symposium on VLSI (GLSVLSI) 2017, Banff, Canada, May 2017.

Prior Work Experience

2022–2026 **Silicon SoC Architect**, *Intel*.

2021–2022 **Intern**, *Intel*.

2008–2011 **Senior Software Engineer**, *Huawei Technologies India Private Limited, India*.
Responsible for development and platform support for applications like Festival Calendar, Messaging Application, RSS Feed Reader for Android Smartphones and Feature phones.

Selected Awards

2015-2020 **Study Award**, *NUS Strategic India Initiative*.

2015-2020 **Fellowship**, *Visvesvaraya PhD Scheme for Electronics & IT, Government of India*.

2013 **Appreciation Letter**, *Dean (Academic Program), Student Mentoring Program*.

2019 **Excellence in Teaching Assistantship**, *IIT Bombay*.

Technical Skills

Simulators **Sniper, booksim**.

Languages **C, C++, Python, Verilog, VHDL**.

Technical Services

2017 **Sub Reviewer**, *International Symposium on VLSI Design and Test (VDAT)*.

2015 **Organization Team Member**, *Asian Test Symposium (ATS)*.

2014 **Organization Team Member**, *International Conference on VLSI Design (VLSID)*.

Note: Last updated on 20th Jul 2026